

# Response to Editor and Reviewers

## **Causal WeatherGraph: screen-confirm-replicate discovery of lagged wind-humidity-cloud dependencies in global atmospheric reanalysis**

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Xinchen Geng · University of Southern California

gengx@usc.edu · 8 October 2026

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Dear Editor and Reviewers,

Thank you for the detailed assessment of my manuscript. I have rewritten the paper around a screen-confirm-replicate framework and revised the interpretation of the atmospheric dependencies. The responses below explain each change, identify its location in the revised manuscript, and state the limitations that remain.

1. The revised framework separates own-history screening, confirmation under all observed regional histories, and replication in two held-out periods. Of 2,574 candidate-lag hypotheses, 2,187 pass screening, 1,550 pass confirmation, and 760 and 667 replicate in the two respective periods. Unconfirmed hypotheses provide a separate comparison group.
2. The inference assessment now includes calendar-time HAC calculations, residual diagnostics, known-null and known-structure simulations, symmetric direction tests, and repeated spatial reference graphs. The manuscript states the remaining calibration and identification limitations and revises the earlier claims about chain enrichment, seasonal strengthening and cloud regimes.
3. The late ERA5 segment was reacquired at native resolution and conservatively regridded. Same-date route comparisons, separate temporal evaluation segments, and CERES cloud-substitution tests now document acquisition-route and product dependence.
4. A pre-specified wind-humidity eddy convergence analysis tests the predictive contribution of the quadratic moisture-flux term. The reported cloud-prediction gains are small, and the response distinguishes these gains from atmospheric intervention effects.
5. The presentation has been reorganized, the bibliography has been revised, and clean and marked manuscripts, Supplementary Material, machine-readable Supplementary Data and reproducible source files accompany the revision. The editorial response explicitly explains the shared-author citation exceptions.

I have retained unfavorable findings and outstanding validation limits in the revised text. The point-by-point responses below indicate where a request is addressed by new evidence, by a narrower interpretation, or by an explicit limitation.

Sincerely,  
Xinchen Geng

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This document answers six editorial requirements and all reviewer comments. Reviewer 1's grouped report is divided into 44 subitems in its original order; Reviewer 2 has five numbered comments and Reviewer 3 has seven. The subitem codes are navigation aids.

Reviewer comments are shown in shaded text, followed by my response and the location of the corresponding revision. Section, figure and table numbers refer to the clean revised manuscript; tables beginning with S refer to the Supplementary Material. The marked manuscript accompanies the clean version.

The responses use the final revision with native ERA5 conservative regridding and the wind-humidity eddy convergence analysis. Older intermediate response drafts and their page numbers are superseded.

# Response to the Editor

## ED-1 · Reference count and citation concentration

### COMMENT

Please ensure that the total number of references does not exceed 50. In addition, self-citations and citations to any single journal or author should be limited to a maximum of three, except in clearly justified special cases.

### RESPONSE

The revised manuscript contains 33 references, and no single journal contributes more than three. I checked author identities using the full bibliographic metadata rather than counting surnames alone. There are no self-citations to Xinchen Geng.

Two shared authors, Alvaro Sanchez-Gonzalez and Peter Battaglia, each appear in four cited papers: GraphCast, NeuralGCM, GenCast and WeatherBench 2. I request the permitted, narrowly justified exception for these two identities. These papers support distinct statements about deterministic graph-based prediction, hybrid physical/neural modelling, probabilistic ensemble forecasting and a standardized evaluation framework. Each is cited at the relevant statement; retaining all four provides the specific forecasting and evaluation context against which the contribution of this paper is defined. I have made this exception explicit rather than claim that every author occurs at most three times.

Revised manuscript: Sections 1 and 2.1; References.

## ED-2 · Individual citations at the supporting statements

### COMMENT

Please avoid citing references in large blocks. For example, citation formats such as [1,2], [1-4], or [1,2,3] should be avoided. Retain only the most relevant references and cite them individually where they directly support the specific statement.

### RESPONSE

I have revised the citation-bearing sentences so that references are attached individually to the statements they support. The current manuscript has no multi-key citation commands. Forecasting frameworks, causal-discovery methods and atmospheric evidence are introduced with separate statements describing the relevant result or capability. The clean PDF uses individual numerical citations; adjacent grouped citations and compressed ranges are not used as a substitute for this restructuring.

Revised manuscript: Sections 1-5 and the corresponding individual citations throughout the manuscript.

### ED-3 · Reference numbering in order of first citation

#### COMMENT

Please order the references according to their first citation sequence in the manuscript, rather than alphabetically.

#### RESPONSE

The references are numbered in order of first citation using the numerical elsarticle-num bibliography style. I checked the first occurrence of each citation in the completed source against the order of the 33 entries in the compiled bibliography. This check uses the actual manuscript and bibliography output, including citations in captions, rather than the storage order of entries in the bibliographic database.

Revised manuscript: References and numerical citations throughout the clean revised manuscript.

### ED-4 · Highlight length

#### COMMENT

Please ensure that each highlight is within 80 characters.

#### RESPONSE

The five highlights have been rewritten to reflect the revised framework and findings. Their displayed lengths are 77, 77, 79, 76 and 78 characters, respectively, including spaces and punctuation. All five therefore meet the 80-character limit. The separate Highlights file contains the same text as the manuscript.

Revised manuscript: Highlights page and the separate Highlights file.

### ED-5 · English revision and marked changes

#### COMMENT

Please conduct a thorough English proofread to improve the overall clarity, grammar, and presentation of the manuscript. Please track all edits made during the revision process.

#### RESPONSE

I have revised the English throughout the manuscript, including the title, abstract, problem formulation, interpretation of results and limitations. The terminology now consistently distinguishes conditional lagged predictive dependence from stronger causal interpretations. Repeated background material has been shortened, definitions and counting units have been clarified, and the discussion has been reorganized around the evidence that does and does not replicate.

I provide both a clean manuscript and a marked manuscript generated from the retained original local source and the revised source. The marked version displays deletions and additions, with the figures and bibliography needed to read those changes. This response reports the language revision undertaken; it does not claim certification by an external language-editing service.

Revised manuscript: Entire clean manuscript and the accompanying marked manuscript.

**ED-6 · Relevance of the retained references****COMMENT**

Please carefully check the relevance of all references. If any reviewer requested citations that are not directly relevant to the topic of the paper, please do not include them.

**RESPONSE**

I have reviewed the bibliography against the claims, data products and methods actually used in the revised paper. The retained references support atmospheric data provenance and cloud controls, time-series causal discovery, the comparison methods, statistical inference, replication, and the physical interpretation of the added diagnostics. The revised Related Work section and positioning table explain the relationship to the methods raised by the reviewers.

References have not been added solely to satisfy a citation request. Recent relevant work has been incorporated, while the original methodological sources for the Benjamini-Hochberg procedure, PCMCI and Regime-PCMCI are retained because those procedures are used or directly compared. The revised bibliography contains 33 entries, of which 30 were published in 2021 or later.

Revised manuscript: Sections 1, 2 and 3; Table 1; References.

# Response to Reviewer 1

The comments below follow the sequence of the reviewer's attached report. The section letters in the identifiers retain the grouping used for the response.

## R1-A1 · Mean pathway lag in the abstract

### COMMENT

the approximately 24-hour mean total pathway lag;

### RESPONSE

I have added the approximately 24-hour summed path lag to the abstract together with its main qualification: it follows the tested lag window. Section 4.5 now reports 3,777 screened all-regime chains and a mean summed lag of 3.99 six-hourly steps, or approximately 23.9 hours. Section 3.9 defines the averaging unit as a lag-resolved two-link chain and explains that uniform detection over lags 1-3 already gives an expected summed lag of four steps.

The joint-window analysis and known-lag simulations show why this descriptive statistic cannot identify a transit time. Information remains distributed across the 1-3, 4-8 and 9-12 source-lag windows. I therefore retain the requested number without treating it as a measured physical propagation time.

Revised manuscript: Abstract; Sections 3.9 and 4.5; Figure 6; Supplementary Table S10.

## R1-A2 · Partial humidity adjustment in the abstract

### COMMENT

the fact that humidity explains only part of Wind → Cloud Cover;

### RESPONSE

I have added an explicit abstract statement that wind-cloud dependence largely persists after conditioning on intermediate humidity. The revised analysis uses the specified intermediate region and temporally aligns the wind source, humidity intermediate and cloud outcome, rather than interpreting target-region adjustment as mediation. Among 20 strong paths, the direct wind-cloud term still improves held-out prediction for 14 paths after the specified intermediate humidity is included. This count remains 14 after adding six physical controls that precede the path source.

The earlier 3.69% average coefficient attenuation is retained as a historical diagnostic. I describe it as partial, control-sensitive statistical adjustment; neither this percentage nor a product of regression coefficients identifies a causal mediated fraction.

Revised manuscript: Abstract; Sections 3.9, 4.5 and 5.1; Supplementary Tables S10 and S12.

**R1-A3 · Concrete regime finding in the abstract****COMMENT**

one concrete finding about regime dependence;

**RESPONSE**

I have added two concrete regime findings to the abstract: seasonal modulation changes sign between hemispheres, and cloud-state contrasts do not replicate. Section 4.4 now reports direct source-by-state interaction tests. Concurrent and antecedent cloud states yield 173 and 124 threshold-passing interactions among 2,574 discovery tests, but only 4 and 1 among 2,574 held-out tests. This comparison replaces an interpretation based solely on differences between separately thresholded graph counts.

I also define winter locally in each hemisphere. The held-out mean winter-minus-summer wind-to-humidity slope is +0.0084 in the Northern Hemisphere and -0.0147 in the Southern Hemisphere. These signed conditional-slope contrasts do not support uniform global winter strengthening.

Revised manuscript: Abstract; Sections 3.8 and 4.4; Figure 5; Supplementary Table S7.

**R1-A4 · Spatial background in the abstract****COMMENT**

the persistence of substantial spatial-background dependence in matched controls; and

**RESPONSE**

I have made the spatial-background qualification explicit in the abstract. In the revised full-record analysis, 85.7% of the 2,574 physics-guided edge-lag hypotheses pass the stated threshold, compared with a mean of 81.3% across 100 distance-matched random candidate sets. The roughly four-percentage-point excess is modest relative to the high background rate.

The graph contains 3,733 lag-resolved wind-humidity-cloud chains against a reference mean of 3,871, with an upper-tail reference probability of 0.81. The symmetric comparison likewise provides no chain-count enrichment. I therefore separate modest edge-level specificity from chain abundance and do not interpret the latter as independent evidence of an atmospheric mechanism.

Revised manuscript: Abstract; Sections 3.11, 4.3 and 5.1; Figure 4; Supplementary Table S14.

**R1-A5 · Observational predictive interpretation****COMMENT**

a clearer statement that the reported discoveries are lagged observational predictive dependencies, obtained largely through linear methods and subject to spatial and temporal confounding.

**RESPONSE**

I have made the estimand explicit: each real-data edge represents a conditional linear predictive coefficient under a stated information set. The abstract concludes that the recovered structure consists of distributed conditional linear predictive dependencies rather than interventional effects, and its opening identifies shared history, spatial background and test miscalibration as central risks. The framework name refers to Granger-type prediction, as explained in the Introduction.

Full conditioning and temporal replication strengthen the evidence for reproducible predictive relationships, but they cannot remove unobserved confounding. The hidden-driver simulations demonstrate this limitation directly. I also retain the finite-sample HAC calibration caveat and avoid interpreting the thresholded graph as a generally validated causal graph or a guarantee of 5% error control.

Revised manuscript: Abstract; Sections 1, 3.1, 4.1, 5.4 and 6.

**R1-I1 · Focused atmospheric and causal-discovery literature****COMMENT**

add recent, topic-specific literature on space-time causal discovery and cloud-controlling factors;

**RESPONSE**

I have reorganized the literature review around the specific discovery and atmospheric questions addressed by the revision. Section 2.2 discusses time-series causal discovery, nonstationarity, regime learning and spatial stencils, including Runge (2023), Chen (2024), Docquier (2024), CausalTime (2024) and CaStLe (2025), while distinguishing these methods from the proposed verification workflow.

Section 2.3 now discusses cloud-controlling factors, vertical thermodynamic structure, moisture convergence and reanalysis cloud biases. The added discussion includes Andersen (2023), Wilson Kemsley (2024), Heitmann (2024), Tootoonchi (2025) and Hellmuth (2025). I use these studies to motivate controls and limitations; results for high-cloud radiative responses or cloud liquid water are not presented as direct validation of the total-cloud-cover graph.

Revised manuscript: Sections 1, 2.2 and 2.3; Table 1.

## R1-I2 · Atmospheric rationale and vertical support

### COMMENT

strengthen the atmospheric rationale for scalar wind speed, 850-hPa variables, and column cloud cover;

### RESPONSE

I have explained that scalar wind speed measures circulation intensity but carries no transport direction, that 850-hPa variables sample lower-tropospheric conditions, and that total cloud cover integrates a different vertical support. The revision also acknowledges that 850 hPa is below ground in some high-terrain cells. Along-connection wind and moisture-flux projections, six additional atmospheric controls, and explicit horizontal-transport diagnostics address selected aspects of this mismatch.

The two-footprint transport experiment improves humidity-change prediction by 11.02% with ridge and 12.51% with boosting, while both cloud intervals include zero. The separate pre-specified eddy-convergence test yields small midlatitude cloud gains of 0.456% and 0.359% in the two held-out segments. I interpret these as conditional predictive contributions of resolved horizontal transport, without claiming a closed column budget or an identified cloud-formation mechanism.

Revised manuscript: Sections 2.3, 3.2, 3.9, 3.10, 4.5, 4.7 and 5.2.

## R1-I3 · Earlier reanalysis limitations and independent cloud

### COMMENT

discuss ERA5/reanalysis-specific limitations earlier;

### RESPONSE

I have moved the reanalysis qualification into the Introduction and expanded it in the data section. ERA5 fields combine a forecast model with assimilated observations; total cloud cover is a model diagnostic and can inherit cloud-physics biases and dependence on humidity. The revision now checks cloud-related relationships with CERES hourly satellite cloud over 2017-2025, rather than relying only on the earlier two-region annual comparison.

I also corrected the acquisition-route discontinuity by replacing the coarse interpolated CDS continuation with native fields processed by the same conservative regridding as WeatherBench 2. Same-date comparisons cover 152 timestamps. Satellite agreement and matched acquisition routes improve the provenance evidence, but neither establishes an error-free cloud target nor identifies the causal graph. The shorter CERES segment does not support the direction asymmetry.

Revised manuscript: Sections 1, 2.3, 3.2, 4.6 and 5.4; Figure 7; Supplementary Tables S1 and S15.

**R1-I4 · Narrowed novelty claim****COMMENT**

narrow and sharpen the novelty claim;

**RESPONSE**

I have narrowed the contribution to a screen-confirm-replicate workflow for checking directed dependencies in strongly dependent regional records. The innovation claimed is the combination of physics-guided candidate selection, full-history confirmation, temporally separated replication, negative controls and explicit provenance checks. I do not claim a new universal identification theorem or that the method discovers a previously established atmospheric mechanism.

The benchmark discussion now states that all-history regression and PCMCi have comparable recovery accuracy in the tested settings, and that Regime-PCMCi provides hidden-state learning absent from prescribed regimes. The additional eddy-convergence experiment contributes a pre-specified test of a physically interpretable nonlinear predictor. Its small predictive gains remain distinct from an interventional effect or general superiority over existing methods.

Revised manuscript: Sections 1, 2.2, 4.8, 5.3 and 6; Tables 1 and 9.

**R1-I5 · Primary and secondary objectives****COMMENT**

explicitly state one primary objective and several secondary objectives at the end of the introduction; and

**RESPONSE**

I have stated a single primary objective in the Introduction: identify regional wind-humidity-cloud dependencies that survive conditioning on all observed regional histories and recur in held-out years. The secondary objectives address spatial background, imposed direction, endogenous regime definitions and lag-window sensitivity; satellite substitution for reanalysis cloud; and the predictive contribution of the quadratic wind-humidity component of moisture flux.

The following framework description and contribution list use the same hierarchy, so the main empirical output is the replicated conditional dependency set. These objectives describe the revised study. I distinguish them from the separately frozen eddy-convergence test and do not characterize the original full-record investigation as prospectively registered.

Revised manuscript: Section 1, paragraph beginning with the central research question and the contribution list.

## R1-I6 · Hypotheses, validation and exploration

### COMMENT

distinguish hypotheses, validation analyses, and exploratory analyses.

### RESPONSE

I have added an explicit analysis-role table. The three-stage analysis, symmetric direction comparisons and pre-specified eddy-convergence test are distinguished from validation exercises using known structures, null graphs, satellite substitution and acquisition-route comparisons. Regime interactions, hemispheric contrasts, path timing and transport diagnostics are marked as exploratory.

The manuscript also separates a frozen revision protocol from prospective validation. Although discovery preprocessing and selection are fixed before replication, the later years had been examined in descriptive analyses. I therefore call this a retrospective temporal holdout. The term confirmatory in the role table denotes the specified analysis procedure; it does not establish exact finite-sample calibration, causal identification or a previously untouched test dataset.

Revised manuscript: Sections 3.7, 3.10 and 3.11; Table 3; Section 5.4.

## R1-M1 · Dependence-aware inference and residual diagnostics

### COMMENT

The key weaknesses concern dependence-robust inference, incomplete bootstrap and null-model specifications, incomplete reporting of sampling and regime sizes, unclear multiple-testing scope, and the absence of an independent holdout validation.

### RESPONSE

I have replaced the original nested F-test as the main inferential procedure with calendar-time HAC inference, using a Bartlett kernel and a primary bandwidth of 64 six-hourly steps, with 32 and 128 as sensitivity values. Regime-excluded scores remain at their calendar positions. Residual diagnostics now show substantial daily dependence, and a month-by-UTC-hour anomaly sensitivity reduces the median 24-hour residual correlation from 0.263 to 0.099.

The known-null simulations also disclose remaining miscalibration: at length 16,384, the heteroskedastic-null empirical FDR is 0.100 with HAC, with a 95% interval of 0.055-0.174. I therefore treat adjusted values as operational thresholds and rely on replication and negative controls for additional evidence, without claiming general nominal FDR control.

Revised manuscript: Sections 3.5, 3.6, 3.11, 4.1 and 5.4; Figure 2.

## R1-M2 · Bootstrap design and scope

### COMMENT

incomplete bootstrap and null-model specifications

### RESPONSE

I have replaced the small legacy stability exercise with a specified bootstrap of the lagged calendar design. Lagged design rows are constructed before blocks are sampled, preserving their original temporal relationships rather than creating artificial lags where sampled blocks meet. The analysis uses block lengths of 64, 128 and 256 six-hourly steps and 200 replicates for each setting.

The reported sensitivity covers twelve fixed edge-regime models. Supplementary Table S8 presents their coefficient intervals and sign stability across block lengths. I explicitly limit the interpretation to this diagnostic subset: it is not an all-edge re-selection bootstrap, a graph-wide stability guarantee, or evidence that the nominal HAC/BH error rate has been established. Legacy stability summaries are identified separately in the supplement.

Revised manuscript: Section 4.9; Supplementary Table S8.

## R1-M3 · Random graph design and mixing

### COMMENT

incomplete bootstrap and null-model specifications

### RESPONSE

I have specified the randomization constraints and repeated the candidate-graph comparison 100 times, preserving variable types, candidate counts and great-circle distance strata and recomputing BH within each randomized candidate set. The reported reference distributions distinguish edge-level threshold rates from lag-resolved chain counts; the revised physical graph has no total-chain enrichment.

The degree-preserving alternative is now reported with its feasibility and mixing limitations. On the original symmetric support, the full-record constraints admit only three graphs, and the two-step chain count is fixed by the intermediate-node degrees. Expanded-support chains change few edges and do not mix adequately. I therefore do not use this procedure as a calibrated inferential null for chain enrichment.

Revised manuscript: Section 3.11; Section 4.3; Figure 4; Supplementary Table S14.

## R1-M4 · Sampling, regimes and effective response counts

### COMMENT

incomplete reporting of sampling and regime sizes

### RESPONSE

I have distinguished raw timestamps, eligible responses and edge-lag tests. The aligned record contains 68,668 six-hourly timestamps. After the three initial lags, discovery uses 58,437 responses; the primary and second replication segments contain 5,884 and 4,344 responses, respectively. Held-out predictors may reach into the preceding segment, while preprocessing parameters remain fixed from discovery.

Table 6 now reports actual full-record response counts for all seven descriptive regimes alongside their common 2,574-test denominator. For example, high humidity has 13,731 responses and high cloud has 13,734, whereas the all-time regime has 68,665. The methods explain the overlapping masks and their response-time definition, so these regimes are not interpreted as independent samples.

Revised manuscript: Sections 3.1-3.3, 3.7 and 3.8; Tables 2 and 6; Supplementary Data S1 and S3.

**R1-M5 · Explicit multiple-testing families**

**COMMENT**

unclear multiple-testing scope

**RESPONSE**

I have made the multiple-testing family explicit for each analysis. Stage 1 and Stage 2 each adjust across all 2,574 candidate-lag hypotheses. Stage 3 applies directional tests within the confirmed family, with the unconfirmed hypotheses evaluated separately as negative controls. The descriptive regime screen supplies both the original regime-by-target-variable adjustments and a global 18,018-test sensitivity. The symmetric and eddy-convergence analyses state their own hypothesis counts and families.

These definitions make threshold decisions reproducible, but they do not by themselves prove graph-wide FDR or replicability-error control. The null simulations retain finite-sample departures from nominal calibration, and spatially dependent direction comparisons are treated descriptively. I avoid transferring a BH guarantee across these different families or across the entire analysis collection.

Revised manuscript: Sections 3.5-3.7 and 3.10; Section 4.1; Supplementary Data S1-S3 and S6.

**R1-M6 · Retrospective temporal replication**

**COMMENT**

the absence of an independent holdout validation.

**RESPONSE**

I have replaced selective retesting of full-record strong edges with discovery on 1979-2018 and two temporally separated evaluations. Preprocessing, candidate definitions and stage decisions are frozen, and all 2,574 original candidate-lag hypotheses are represented, including 1,024 unconfirmed negative controls. Of 1,550 confirmed hypotheses, 760 replicate in 2019-01-01 to 2023-01-10 and 667 in 2023-01-11 to 2025-12-31; the corresponding control counts are 95 and 80.

Directional replication refits the full model within each evaluation segment, while predictive gains use discovery-frozen coefficients. I keep those two assessments distinct. Since the later years were previously examined in descriptive analyses, this remains retrospective temporal evaluation; the revision does not claim a pristine prospective holdout.

Revised manuscript: Sections 3.3 and 3.7; Section 4.2; Table 4; Figure 3; Section 5.4.

## R1-M7 · Equations, implementations and reported results

### COMMENT

The validity of the equations and the results related to it, should be verified properly before final submission.

### RESPONSE

I have made the principal estimands and computations explicit, including the two information sets, anomaly transformation, regression and HAC equations, BH family definition, frozen predictive gain, time-aligned path model and moisture-flux decomposition. Definitions now distinguish incremental  $R^2$  from partial  $R^2$  and explain the calendar alignment of lagged variables and scores.

For the eddy-convergence calculation, the manuscript reports numerical conservation checks: the global area integral is approximately  $5 \times 10^{-17}$  of the summed absolute flux, and explicit boundary-face summation agrees with regional values to approximately  $4 \times 10^{-16}$ . These checks support the stated implementation. They do not validate causal identification or establish every statistical assumption, and I do not represent numerical agreement as proof of nominal error control.

Revised manuscript: Sections 3.1 and 3.3-3.10; Section 4.7; Supplementary Data S6.

## R1-P1 · Figure readability at publication size

### COMMENT

Make all axes, legends, color scales, and units readable at journal-column width.

### RESPONSE

I have reorganized the main figures around the revised framework and its principal comparisons, using vector PDF graphics with explicit axes, legends and units. The plots now separate calibration, temporal replication, direction and null comparisons, regimes, timing, satellite substitution and eddy-convergence gains rather than relying on dense graph counts alone.

A check of the assembled PDF nevertheless identifies a remaining presentation limitation: some framework labels and compact multi-panel annotations become small at the current placement size. I therefore do not claim that every label is already readable at journal-column width. The framework and regime figures still require a final typography and production-size check before submission.

Revised manuscript: Figures 1-8, particularly Figures 1 and 5.

## R1-P2 · Definition of edge-counting units

### COMMENT

Define whether “edge count” means edge-lag tests, unique directed edges, or unique region-variable edges.

### RESPONSE

I have defined the counting units explicitly in the problem formulation. A candidate edge is one directed region-variable pair; an edge-lag hypothesis adds a specified lag; and an edge-lag-regime test adds a specified regime. A unique directed edge collapses accepted lags, whereas a lag-resolved chain records both links and both lags.

The results now retain these distinctions. There are 858 original candidates and 2,574 candidate-lag hypotheses, while seven overlapping descriptive regimes produce 18,018 tests. In the primary evaluation, 760 replicated hypotheses correspond to 450 unique directed edges. I report both numbers so that a multi-lag relationship cannot be mistaken for several independent spatial connections or several independent atmospheric mechanisms.

Revised manuscript: Section 3.1; Section 3.4; Section 4.2; Table 4.

## R1-P3 · Consistent significance and edge terminology

### COMMENT

Use consistent terminology for “significant edges,” “significant hypotheses,” and “unique edges.”

### RESPONSE

I have tied the terminology to defined objects and stage rules. Screened hypotheses pass the own-history threshold; confirmed hypotheses also pass the full-history stage; replicated hypotheses pass the directional temporal rule within the confirmed family. Unique edges are explicitly deduplicated over lags. This prevents a threshold count, a stage count and a unique-edge count from being treated as interchangeable quantities.

Some tables retain the compact label significant. Section 4.1 and the limitations clarify that this means passing the stated operational threshold, since exact finite-sample calibration is not established. I do not use that label to imply interventional causality or a general graph-wide false-discovery guarantee.

Revised manuscript: Sections 3.1 and 3.5-3.7; Sections 4.1-4.3; Supplementary Data descriptions.

## R1-P4 · Sample sizes for regime comparisons

### COMMENT

Report sample size in every regime figure.

### RESPONSE

I have made the sample support for the regime comparisons visible in the results. Table 6 reports the exact eligible response count for each full-record regime. Figure 5 identifies the discovery and evaluation periods, gives shared response counts for the hemispheric contrasts, and states the test denominators of 2,574 per state and 1,287 per hemisphere. This separates the number of observations from the number of tested coefficients.

The current figure does not repeat every model-specific response count in every panel; the detailed regime counts remain in Table 6 and the accompanying methods. I therefore avoid implying that all panels use one common nominal sample size, especially where masks and local seasons differ.

Revised manuscript: Figure 5; Section 4.4; Table 6; Section 3.8.

## **R1-P5 · Rates and uncertainty in regime comparisons**

### **COMMENT**

Avoid comparing raw counts across regimes without showing rates or uncertainty.

### **RESPONSE**

I have stopped interpreting raw differences between separately selected regime graphs as direct evidence of regime dependence. Table 6 gives response counts, a common 2,574-test denominator and relevant edge-type shares. The main comparison instead estimates source-by-state interactions and local winter-minus-summer slope contrasts, with calendar-time HAC uncertainty.

For the hemispheric comparison, influence scores are combined on the same calendar, retaining covariance between hemispheres. The held-out wind-to-humidity contrast is 0.0231 with a 95% interval of 0.0178-0.0284. These are signed conditional-slope contrasts, not changes in absolute strength or causal difference-in-differences estimates. Because graph tests are dependent, I do not attach independent-binomial confidence intervals to their threshold counts.

Revised manuscript: Sections 3.8 and 4.4; Table 6; Figure 5; Supplementary Table S7.

## **R1-P6 · Numerical precision**

### **COMMENT**

Use consistent decimal precision.

### **RESPONSE**

I have used consistent precision within comparable table columns: counts are integers, rates generally have three decimal places, and predictive gains retain enough digits to show their small scale. Scientific notation is used where ordinary rounding would hide numerical provenance differences or conservation checks. For example, the main replication table consistently reports rates such as 0.490 and 0.430.

The machine-readable files retain the underlying numerical values. Display rounding does not determine threshold membership, and the extra digits supplied for reproducibility are not presented as additional statistical certainty.

Revised manuscript: Tables 2-9; Supplementary Tables S1-S17 and machine-readable data.

## R1-P7 · Averaging units for lag summaries

### COMMENT

Explain whether mean lags are calculated over significant hypotheses, unique edges, or chains.

### RESPONSE

I have specified that the approximately 23.9-hour mean is calculated over lag-resolved two-link chains. If one spatial route is represented by several lag combinations, each combination contributes to that mean; it is not a mean over deduplicated spatial routes. The separate long-window screen reports single-edge lag summaries, which are a different quantity.

For the joint-source models, Section 3.9 defines an absolute-coefficient-weighted centroid across all included source lags. Its values of 2.01, 3.57 and 6.44 steps correspond to maximum windows of 3, 6 and 12 steps. The manuscript uses this sensitivity to limit interpretation, rather than combining these different averages or treating any of them as an air-parcel transit time.

Revised manuscript: Sections 3.9 and 4.5; Figure 6; Supplementary Table S10.

## R1-P8 · Explicit denominators

### COMMENT

Include the exact denominator whenever a fraction or ratio is reported.

### RESPONSE

I have supplied the counting bases for the main rates and comparisons. Replication is reported as 760 of 1,550 confirmed hypotheses in the primary period and 667 of 1,550 in the second, with controls of 95 and 80 among 1,024 unconfirmed hypotheses. Direction comparisons use 636 matched pairs per variable direction, path diagnostics use 60 selected paths with 20 per stratum, and band-level eddy gains average 22 regions.

The accompanying definitions distinguish response counts, tested hypotheses, unique edges, paths and regions. A rate within a selected path stratum is not interpreted as the prevalence of atmospheric pathways, and a repeated graph draw is not treated as a new independent observational dataset.

Revised manuscript: Sections 3.1, 4.2-4.7; Tables 4-8; Supplementary Data.

## R1-P9 · Analysis status in presentation

### COMMENT

Clearly mark exploratory versus confirmatory analyses.

### RESPONSE

I have added a dedicated table assigning each analysis its role. The three-stage procedure, symmetric direction comparisons and frozen eddy-convergence test are listed separately from validation simulations, null-graph comparisons, satellite substitution and acquisition-route checks. Regime, timing, transport and robustness diagnostics are explicitly exploratory.

I also state two qualifications needed to interpret these labels. The temporal split is retrospective because evaluation years appeared in earlier descriptive work, and the HAC/BH thresholds do not establish exact error control. The eddy-convergence design was fixed before its results, including the two endpoints and the control rule, but this does not convert all earlier analyses into confirmatory tests or make the observational estimand causal.

Revised manuscript: Table 3; Sections 3.7, 3.10 and 3.11; Section 5.4.

## R1-P10 · Complete machine-readable edge results

### COMMENT

Provide supplementary machine-readable tables containing all edge-level results, including region IDs, variables, lags, coefficients, p-values, q-values, partial R<sup>2</sup>, and stability.

### RESPONSE

I have provided Supplementary Data S1 for all 2,574 original candidate-lag hypotheses, with source and target identifiers, variables, lag, distance, stage coefficients, standard errors, p- and q-values, partial R<sup>2</sup>, held-out results and stage indicators. Additional files cover all 3,816 symmetric hypotheses per period, the 18,018 regime-screening tests, cloud substitution and eddy-convergence analyses. Region identifiers and centroids are supplied in S0.

Temporal replication and frozen predictive-gain flags are available for reproducibility assessment. Corrected bootstrap stability, however, covers only twelve fixed diagnostic models in Table S8; the all-edge data do not contain a valid full-pipeline bootstrap score for every edge. I explicitly distinguish this limitation from the completeness of the coefficient and test tables and from legacy stability summaries.

Revised manuscript: Data and code availability; Supplementary Material Section S7; Supplementary Data S0-S6; Supplementary Table S8.

## R1-D1 · Consolidated evidence tables

### COMMENT

consolidate redundant tables;

### RESPONSE

I have consolidated the main presentation around the questions resolved by the revision. Compact tables describe the framework's position, data and analysis roles, three-stage replication, symmetric directions, regimes, satellite substitution, eddy convergence and method benchmarks. The results narrative follows these questions instead of repeating successive robustness tables.

Detailed screening variants remain in the supplement, with their historical data route and inferential scope identified. The main manuscript retains adverse findings that alter the interpretation: incomplete calibration, sign changes after conditioning, absent chain enrichment, nonreplicating cloud-state contrasts and small cloud gains. The consolidation therefore reduces repetition without removing the evidence that limits the principal claims.

Revised manuscript: Tables 1-9; Sections 4.1-4.9; Supplementary Tables S1-S17.

## R1-D2 · Detailed robustness material in the supplement

### COMMENT

move detailed robustness tables to supplementary material;

### RESPONSE

I have moved detailed screening sensitivities and extended diagnostics into a structured supplement. It includes acquisition-route comparisons, historical-period and regionalization checks, regime graph summaries, external-state controls, block-bootstrap diagnostics, falsification controls, path models, known-structure recovery, degree-constrained nulls, satellite agreement and complete eddy-convergence endpoints.

The supplement identifies which tables reproduce the original submission's interpolated continuation and which use the reacquired conservative product. The opening note reports that 670 of 18,018 threshold decisions change when the full-record screen is rerun on the revised inputs. Calibration failure, conditioning sensitivity and limits on causal interpretation remain in the main text; the supplement supplies detail without concealing these central qualifications.

Revised manuscript: Section 4.9; Supplementary Tables S1-S17; Supplementary Material Section S7.

## R1-D3 · Maps, distributions and empirical reference plots

### COMMENT

redesign the main figures around maps, distributions, forest plots, and empirical nulls;

### RESPONSE

I have redesigned the main empirical figures around the revised scientific comparisons. Figure 3 combines stage and replication summaries with the geography of replicated cross-region links. Figure 4 presents matched-direction comparisons and the empirical distributions from repeated distance-matched candidate sets. Figures 5 and 6 show seasonal contrasts, lag-window sensitivity and predictive gains rather than treating dense network diagrams as self-validating evidence.

The satellite and eddy-convergence figures add product comparisons and interval estimates for the main predictive endpoints. The resulting figure set uses maps, reference distributions and effect contrasts to separate reproducibility, specificity and physical interpretation. As noted in response R1-P1, final typography at journal production size remains a separate presentation check.

Revised manuscript: Figures 2-8, particularly Figures 3-6.

## R1-D4 · Effect sizes and uncertainty

### COMMENT

add confidence intervals and effect-size distributions;

### RESPONSE

I have added effect-size summaries and uncertainty alongside threshold counts. Replicated hypotheses have a median held-out partial  $R^2$  of 0.0029 and a median frozen relative MSE gain of 0.27%, showing that numerous selected dependencies can still have modest predictive contributions. Coefficient and test statistics for the complete candidate set are available in the machine-readable data.

Seasonal slope contrasts, transport-loss differences and eddy-convergence endpoints now carry explicit uncertainty definitions. For the two-footprint cloud comparison, the ridge interval spans -0.30% to 0.79% and the boosting interval -1.14% to 0.16%, so neither establishes an improvement. I distinguish these conditional intervals from simultaneous coverage across the graph; unresolved calibration limits the interpretation of real-data HAC intervals.

Revised manuscript: Sections 4.1, 4.2, 4.4, 4.5 and 4.7; Figures 2, 3, 5, 6 and 8; Supplementary Data S1 and S6.

## R1-D5 · Quantified state, latitude, path and null contrasts

### COMMENT

formally quantify regime, latitude, mediation, and null-model contrasts; and

### RESPONSE

I have replaced qualitative graph comparisons with directly defined contrasts. State dependence is assessed through source-by-state interactions; local seasonal contrasts are compared on mirrored hemisphere pairs with shared-calendar covariance.

Latitude-specific predictive gains are reported for the pre-specified eddy-convergence test: midlatitude cloud gains exceed tropical gains by 0.307 and 0.215 percentage points in the two evaluation segments, while polar gains are at least comparable.

For the pathway analysis, I use temporally aligned regressions at the specified intermediate humidity region and repeat them with six preceding physical controls. For spatial specificity, I compare the observed statistics with 100 matched candidate draws. These calculations quantify conditional associations and predictive differences; they do not identify a mediated causal fraction, remove hidden confounding, or make the poorly mixing degree-preserving null valid.

Revised manuscript: Sections 3.8-3.11 and 4.3-4.7; Figures 4-8; Supplementary Tables S7, S12, S14 and S16.

**R1-D6 · Dependence and temporal sensitivity****COMMENT**

strengthen dependence-aware and held-out sensitivity analyses.

**RESPONSE**

I have expanded the sensitivity analysis to calendar-time HAC inference, calendar-design block resampling, discovery-only preprocessing, dense full-history conditioning and two separate temporal replication segments. The month-by-UTC-hour preprocessing sensitivity substantially reduces residual daily dependence while retaining 721 primary-period replications among 1,457 confirmed hypotheses, compared with 117 among 1,117 controls.

The timing analysis additionally fits alternative source windows on aligned responses and evaluates frozen prediction losses; useful information is distributed across 6-72 hours rather than confined to the shortest lags. These comparisons distinguish dependence correction, changes in the conditioning set and temporal prediction. The finite-sample null results remain imperfectly calibrated, and the later years were previously examined, so these additions strengthen sensitivity assessment without establishing a prospective or universally calibrated validation.

Revised manuscript: Sections 3.5-3.7, 4.1, 4.2, 4.5 and 4.9; Supplementary Tables S8 and S11.

**R1-REF1 · Targeted recent references****COMMENT**

Add a targeted set of recent 2024–2026 references.

**RESPONSE**

I have added recent references where they directly support a revised argument or implemented comparison. Examples include WeatherBench 2, NeuralGCM, Heitmann, Wilson Kemsley, Soci, Docquier and CausalTime from 2024, and CaStLe, GenCast, Tootoonchi and Hellmuth from 2025. Together they address data provenance, modern prediction, causal discovery, cloud controls, moisture transport and reanalysis cloud biases.

The current bibliography contains 33 references, rather than the 20 cited in the earlier response draft. I retain older original-method references where the analysis directly uses those methods and avoid adding a 2026 citation solely to fill a date range. The literature discussion states the different targets and assumptions of the cited studies instead of treating recency as evidence for the graph.

Revised manuscript: Sections 1 and 2; Section 3; References.

## R1-O1 · Methodological and educational contribution

### COMMENT

Scientific contribution: Potentially novel and important, primarily as a methodological and reproducibility contribution rather than as a fundamentally new atmospheric mechanism.

Educational value: The paper has meaningful educational value because it demonstrates how to formulate, constrain, validate, and cautiously interpret directed dependency analyses in complex observational systems.

### RESPONSE

I agree that the strongest contribution is methodological and reproducibility focused. The revised manuscript shows why a nearly saturated own-history graph can contain many indirect and common-driver dependencies, how full conditioning changes that graph, and how temporal replication with unconfirmed controls helps assess which conditional relationships recur. The workflow also demonstrates the practical impact of acquisition-route differences on inference.

I have retained negative results and the advantages of competing methods in the settings where they apply. The eddy-convergence result is a small, interpretable predictive contribution, not a newly identified intervention mechanism. The educational value therefore lies in making assumptions, denominators, controls and limitations inspectable across the complete analysis.

Revised manuscript: Sections 1, 4.8, 5.3 and 6.

## R1-T1 · Auditable causal language

### COMMENT

The main concern is not misconduct suggested by the manuscript; it is the need to ensure that the paper's causal language, statistical claims, data provenance, AI disclosure, and computational documentation are fully auditable.

### RESPONSE

I have aligned the principal claims with the stated linear-projection estimand. In the real-data analysis, an edge means that a source lag adds predictive information given a specified set of observed histories. The Introduction explains the Granger-type meaning of causal in the framework name, and the abstract and conclusion explicitly reject an interventional interpretation of the wind-humidity-cloud chain.

The revision further separates temporal replication from causal identification, spatial coherence from mediation, and a lag-window summary from a travel time. The hidden-driver benchmark and discussion make clear that full conditioning on observed variables cannot exclude unobserved common forcing. These definitions allow the causal wording to be checked against the relevant calculation rather than inferred from the graph's arrows.

Revised manuscript: Sections 1, 3.1, 5.4 and 6; Abstract.

**R1-T2 · Auditable statistical claims and unfavorable results**

**COMMENT**

statistical claims

**RESPONSE**

I have retained the results that limit the statistical interpretation, including finite-sample HAC miscalibration, poor own-history structural precision, coefficient-sign changes under full conditioning, the absence of chain-count enrichment and nonreplicating cloud-state contrasts. The cloud-transport intervals include zero, and the smaller second CERES segment does not support the humidity-cloud direction asymmetry; both qualifications are visible in the main results.

Stage rules, hypothesis families, response counts and complete candidate-level statistics are documented so that readers can reconstruct reported decisions. I distinguish descriptive threshold counts from predictive effect sizes and temporal replication, and I do not present BH adjustment or numerical reproducibility as a general proof of nominal error control.

Revised manuscript: Sections 4.1-4.8 and 5.4; Supplementary Data S1-S6.

**R1-T3 · Data provenance and acquisition correction**

**COMMENT**

data provenance

**RESPONSE**

I have documented the ERA5 sources, units, time coverage, anomaly transformation and acquisition routes. The revised 2023-2025 continuation uses native 0.25-degree CDS fields regridded conservatively, matching the WeatherBench 2 route used earlier. On 152 identical timestamps, this removes the discrepancy introduced by directly requested coarse-grid interpolation; regional standardized differences are below  $3 \times 10^{-5}$  with correlations of 1.000.

The supplement preserves the earlier-route comparisons and labels original-submission robustness tables accordingly. CERES now provides a processed 2017-2025 satellite comparison and cloud-substitution tests, rather than a statement that downloads alone validate the results. The available source manifests and machine-readable comparisons support provenance and reproducibility, while the discussion retains the limitations of reanalysis, satellite retrievals and observational dependence.

Revised manuscript: Section 3.2; Section 4.6; Data and code availability; Supplementary Tables S1 and S15; Supplementary Data S5.

## R1-T4 · Transparent AI disclosure

### COMMENT

AI disclosure

### RESPONSE

I have expanded the declaration to identify both Claude and OpenAI Codex and the tasks for which assistance was used. The statement covers language editing, code review, analysis scripting, figure preparation, code development and execution of revision analyses. It therefore does not describe the assistance as proofreading alone.

The declaration assigns responsibility for the text, code, analyses and references to the author; an AI system is not listed as an author. Its statements about author review must reflect the review actually completed before submission. Automated numerical checks and drafting assistance cannot substitute for that responsibility or independently certify that every scientific claim is correct.

Revised manuscript: Declaration of generative AI and AI-assisted technologies in the writing process.

## R1-T5 · Computational documentation and reproducibility

### COMMENT

computational documentation

### RESPONSE

I have linked the analysis code and configurations and provided machine-readable results for the complete screening, confirmation and replication procedures. The supplementary index names each dataset, its scope and identifiers, including region centroids, symmetric comparisons, acquisition-route checks and eddy-convergence outputs. The latter also includes its frozen design file and hash, allowing its stated decision rule to be distinguished from later exploratory interpretation.

Runtime and memory reporting now accompanies the main framework and selected benchmarks, with the workstation identified. These measurements describe the recorded configurations, not a hardware-independent ranking of algorithms. I also distinguish the original-submission robustness outputs from the revised conservative-route analyses so that readers can identify the appropriate data and result version.

Revised manuscript: Data and code availability; Section 4.8; Supplementary Material Section S7; Supplementary Data S0-S6.

## R1-L1 · More direct language

### COMMENT

more direct

### RESPONSE

I have rewritten the manuscript around explicit questions, defined comparisons and numerical results. The Introduction states the principal objective directly, and each results subsection begins with its substantive finding. For example, the direction section distinguishes the modest humidity-cloud asymmetry from the unsupported wind-humidity orientation, while the pathway section states that lag summaries depend on the search window.

General descriptions of strong or robust evidence have been replaced where possible by the relevant denominator, effect size and limitation. This makes the contribution easier to assess without requiring readers to infer what a dense graph or a high significance rate establishes.

Revised manuscript: Throughout, particularly Sections 1, 4 and 6.

## R1-L2 · Reduced repetition

### COMMENT

less repetitive

### RESPONSE

I have consolidated repeated method descriptions into the methodological sections and used the analysis-role table to explain how the individual experiments relate. The results now follow distinct questions: calibration, replication, direction and specificity, regimes, timing and transport, independent cloud, eddy convergence and benchmarking. Detailed screening sensitivities are collected in the supplement.

I retain short qualifications beside the results they constrain, such as the retrospective nature of temporal evaluation and the absence of a causal mediation estimate. This necessary repetition prevents a result from appearing stronger when read on its own, while avoiding repeated inventories of the same robustness procedures.

Revised manuscript: Sections 2-5; Tables 1-3; Supplementary Material.

## R1-L3 · Consistent caution in causal interpretation

### COMMENT

more cautious in its causal language.

### RESPONSE

I have made the main causal qualification consistent across the abstract, estimand definition, discussion and conclusion. The empirical output is a set of conditional lagged predictive dependencies, evaluated under specified linear models and observed information sets. Neither a directed arrow, persistence across periods, satellite agreement nor an added transport feature is treated as an intervention effect.

The manuscript also states that humidity adjustment is not an identified mediated proportion and that approximately 24-hour path lags are not measured transit times. The final conclusion distinguishes the reproducible dependency set and its verification procedure from a uniquely identified wind-humidity-cloud causal chain, while retaining hidden-driver and model-specification limitations.

Revised manuscript: Abstract; Sections 1, 3.1, 5 and 6.

## Response to Reviewer 2

### R2-1 · Serial dependence and multiplicity

#### COMMENT

(1) The validity of significance testing under strong temporal dependence requires further assessment. Section 3.7 applies conventional OLS-based F-tests while controlling only three target-history lags in the principal model. Including these lags does not establish that residual autocorrelation and heteroskedasticity have been adequately addressed. Year-block bootstrap intervals for selected strong edges cannot establish calibration of the full discovery procedure. Please report residual diagnostics and reassess significance using inference procedures appropriate for serial dependence. Simulations with known dependency structures should evaluate the empirical false-discovery rate. Section 3.8 should also distinguish FDR control within each regime-target-variable group from any claim of control across the complete collection of 18,018 tests.

#### RESPONSE

I agree that three target-history lags and intervals for selected edges do not establish valid inference. I have replaced the principal nested F-test with calendar-time Bartlett HAC inference, using a bandwidth of 64 six-hour steps and sensitivity bandwidths of 32 and 128. Excluded regime dates retain their original calendar positions with zero influence scores. The revised analysis explicitly separates the 2,574-hypothesis three-stage family from descriptive regime-by-target-variable families and the global family of 18,018 tests; within-family BH is not presented as full-collection control.

The discovery-period screening residuals have median autocorrelations of 0.263, 0.215 and 0.218 at 24, 72 and 168 hours. Removing the diurnal cycle reduces these to 0.099, 0.046 and 0.027. I also report a calendar-preserving block bootstrap for twelve fixed edge-regime models, with 200 replicates at each of three block lengths. This is a coefficient-stability diagnostic, not validation of the complete selection procedure.

Known-null experiments evaluate all 90 cross-variable lag hypotheses in each simulated dataset. Under heteroskedastic AR(4) noise, empirical FDR is 0.890 for OLS and 0.270 for HAC at length 2,048; at length 16,384 it remains 0.940 and 0.100, respectively, with a 95% interval of 0.055-0.174 for the latter. The externally masked HAC value is 0.070 at that length. These results support using dependence-aware uncertainty but do not establish exact 5% FDR for the real graph. I therefore retain nominal q-values as thresholds, add full-conditioning confirmation and held-out replication with unconfirmed controls, and explicitly acknowledge that replication does not itself prove calibrated full-graph FDR.

Revised manuscript: Sections 3.5-3.7, 3.11, 4.1 and 5.4; Figure 2; Supplementary Table S8 and Supplementary Data S1/S3.

## R2-2 · Imposed directions, saturation and graph nulls

### COMMENT

(2) The candidate graph and nearly saturated discoveries complicate the interpretation of pathway recovery and stability. Section 3.5 explicitly permits Wind→Humidity and Humidity→Cloud while excluding their reverse directions from the principal search space. Consequently, the orientation of the central pathway is partly imposed rather than discovered. Furthermore, Table 11 reports that 857 of 858 candidate edges are significant in at least one regime or lag. Under such saturation, high graph overlap across periods or candidate-neighborhood settings may provide limited evidence of structural specificity. The significant proportions for physical and distance-matched random candidates are also relatively close, at 0.792 and 0.743. Please evaluate bidirectional candidates under comparable spatial constraints and quantify pathway enrichment and graph overlap against repeated null graphs preserving relevant density, degree, and distance characteristics. The manuscript should clearly separate prior-imposed structure from information contributed by the observations.

### RESPONSE

I agree that the original candidate list imposes part of the orientation and that overlap between nearly saturated graphs is weak evidence of structural specificity. I have added a symmetric regional support with identical spatial opportunities for both directions of each wind-humidity, humidity-cloud and wind-cloud relation: 1,272 candidates and 3,816 lag-specific hypotheses.

Under full conditioning, humidity-to-cloud has the larger absolute HAC statistic in 55.2% of 636 discovery comparisons and in 52.2% and 53.6% in the two held-out segments. These are modest descriptive asymmetries because matched pairs remain spatially dependent. Wind-to-humidity is not stronger than its reverse. I therefore distinguish the wind-humidity orientation specified by the prior from the conditional dependencies contributed by the observations.

I also compare the physical candidates with 100 random candidate sets preserving variable types, candidate counts and distance strata, recalculating BH for each set. The full-record crossing fraction is 0.857 versus a random mean of 0.813, but the 3,733 lag-resolved chains are not enriched relative to the null mean of 3,871 (upper-tail probability 0.81). The symmetric-support chain count is likewise not enriched. These candidate references preserve density and distance composition, not individual node degrees.

A separate audit preserves type-specific in/out degrees and distance histograms. Only three full-record graphs and two temporal-overlap graphs are feasible on the original symmetric support. The ordinary chain total is fixed by the intermediate nodes' typed degrees. Expanded-support chains change about 4% of edges and do not mix adequately, so I do not treat their overlap statistics as calibrated null probabilities. The revision consequently retains screening stability as reproducibility evidence and withdraws claims of uniquely recovered orientation or pathway-count enrichment.

Revised manuscript: Sections 3.4, 3.11, 4.3, 4.9 and 5.1; Table 5; Figure 4; Supplementary Table S14.

## R2-3 · Antecedent states and local seasons

### COMMENT

(3) Regime definitions may introduce endogenous selection into the comparisons. Section 3.6 defines humidity and cloud regimes using contemporaneous spatial averages that include the response variable being analyzed. Differences between regime-specific graphs may therefore reflect response-dependent selection or truncation, in addition to genuine changes in atmospheric coupling. Matching sample sizes addresses differences in statistical power but does not resolve this concern. Please include regime definitions based on appropriately preceding information or external circulation classifications, and directly test differences in source effects across regimes. In addition, Sections 4.3 and 5.1 interpret stronger global DJF relationships as stronger dependence under winter conditions. Because DJF corresponds to opposite seasons across hemispheres, this interpretation requires separate Northern and Southern Hemisphere analyses.

### RESPONSE

I agree that matching sample sizes cannot correct selection on the current response. I now describe contemporaneous humidity/cloud regimes as exploratory response-dependent subsets and do not interpret differences between their separately selected graphs as evidence of changing coupling. I fit thresholds on 1979-2018 and add states defined at  $t-4$ , before all three tested source lags. An external alternative uses the last complete monthly Niño-3.4 anomaly before the earliest source, with thresholds fitted on the discovery months.

I directly test source-by-state interaction coefficients, allowing state-specific intercepts and target-history effects and using calendar-time HAC uncertainty. Cloud-state interactions pass BH in 173 concurrent-state and 124 antecedent-state discovery tests, but only 4 and 1 of the 2,574 held-out tests. Humidity-state interactions decline from 117 and 107 to 36 and 37. Adding six antecedent physical controls changes the signs of 30 of 108 high-minus-middle and 24 of 108 low-minus-middle Niño-3.4 contrasts. I therefore report no replicable cloud-state or ENSO modulation in this design, rather than treating the original graph differences as established atmospheric states.

I have also replaced the global-winter interpretation with hemisphere-specific seasons and direct mirrored-region contrasts. Winter is DJF in the Northern Hemisphere and JJA in the Southern Hemisphere. For wind-to-humidity, the signed local winter-minus-summer slope is +0.0036 versus -0.0149 in discovery, and +0.0084 versus -0.0147 held out. The hemispheric contrast is 0.0185 (95% interval 0.0160-0.0210) and 0.0231 (0.0178-0.0284), respectively. Thus, seasonal modulation is supported, but uniform strengthening in winter is not; these are slope contrasts rather than claims of uniformly larger absolute effects.

Revised manuscript: Sections 3.8, 4.4 and 5.1; Table 6; Figure 5; Supplementary Tables S6-S7.

## R2-4 · Path timing, intermediates and transport

### COMMENT

(4) Path composition and lag summaries do not yet establish a propagation or mediation mechanism. Section 3.9 constructs Wind-Humidity-Cloud chains by joining significant edges, but two predictive associations do not by themselves demonstrate transmission through the intermediate variable. The humidity-adjustment diagnostic uses humidity in the target region, whereas the enumerated chains can involve a different intermediate region. It therefore does not directly validate the cross-region pathways being counted. The mean coefficient attenuation of only 3.69% in the all-sample regime also warrants caution. Moreover, mean edge lags near two time steps and chain lags near four time steps are close to what approximately uniform detection over the tested one-to-three-lag range would produce. The shift toward mean lags around six when the maximum lag is extended to twelve raises a related search-window concern. Please evaluate specific intermediate regions with temporally consistent pathway models and assess lag-specific predictive contributions with uncertainty estimates. Physical transport interpretations would also benefit from wind-vector or moisture-flux diagnostics, since scalar wind speed does not encode transport direction.

### RESPONSE

I agree that joining two edges does not identify transmission through humidity. I now evaluate 60 discovery-selected paths, including 20 strong, 20 ordinary and 20 unselected paths, using the specified intermediate region and the aligned sequence  $W_i(t-11-12)$ ,  $H_h(t-12)$ ,  $C_j(t)$ . Among strong paths, adding six physical controls preceding the wind source reduces positive held-out gains for the humidity-to-cloud stage from 19/20 to 17/20; a direct wind-to-cloud contribution remains positive in 14/20. The earlier 3.69% attenuation is retained as a coarse association summary, not a mediated fraction.

I fit source lags jointly in windows of 3, 6 and 12 steps and evaluate deletion of windows 1-3, 4-8 and 9-12 with frozen coefficients. The coefficient centroid shifts from 2.01 to 3.57 to 6.44; a known-lag simulation reproduces window dependence despite a true lag of two steps. All three windows retain held-out predictive information. Principal lag-specific hypotheses have HAC standard errors and frozen-gain uncertainty in Supplementary Data S1; path-stratum summaries remain descriptive and do not supply selection-adjusted mediation intervals.

I also add along-connection wind and moisture-flux projections, conservative moisture-flux convergence, and two fixed Atlantic transport experiments with 500 paired 30-day-block bootstrap draws. Transport features improve subsequent humidity prediction by about 11-13%, but cloud intervals include zero. Separately, a pre-specified wind-humidity eddy-convergence (WHEC) index isolates the quadratic moisture-flux term beyond full conditioning and linear transport. Its pooled midlatitude cloud gains are small, 0.456% and 0.359% in the two held-out segments, while remote and time-shifted controls fail the stated endpoint. These results support limited predictive transport information, not a uniquely oriented causal chain or a physical transit time.

Revised manuscript: Sections 3.7, 3.9-3.10, 4.5, 4.7 and 5.2-5.4; Figures 6 and 8; Table 8; Supplementary Tables S10, S12 and S16-S17; Supplementary Data S1/S6.

## R2-5 · Temporal evaluation and method benchmarks

### COMMENT

(5) PCMCI confirmation is not independent validation, and the methodological contribution needs stronger benchmarking. Section 4.8 selects the strongest stable edges using the complete 1979–2025 record and then evaluates them with PCMCI on the overlapping 2019–2025 segment. The reported confirmation of 50 out of 60 edges demonstrates agreement between procedures on selected relationships, but does not constitute independent temporal validation or establish accuracy across the full graph. Please use an earlier period for discovery and parameter selection, reserving a later period exclusively for evaluation. Validation should include ordinary significant edges and unselected candidates, not only the strongest discoveries. The framework should also be compared with multivariable Granger, PCMCI, and existing regime-dependent approaches under comparable conditions. Known-structure simulations, component ablations, and runtime and memory measurements would help establish the specific benefits of the proposed framework.

### RESPONSE

I agree that PCMCI agreement on 60 full-record-selected edges was not independent validation. I have reorganized the main analysis around discovery in 1979–2018, with preprocessing, candidates and decisions frozen before evaluation in 2019-01-01 to 2023-01-10 and, separately, 2023-01-11 to 2025-12-31. The revised late record uses native 0.25-degree CDS fields conservatively regridded to match WeatherBench 2; a 152-timestamp overlap audit verifies the route. Because the later years were already examined in descriptive analyses, I explicitly call this a retrospective temporal holdout, not a prospective untouched test.

All 2,574 hypotheses are evaluated. Screening retains 2,187, full conditioning confirms 1,550, and 760 replicate in the primary segment versus 95/1,024 unconfirmed controls; 667 replicate in the second segment versus 80 controls. Frozen-coefficient prediction is evaluated separately. PCMCI now covers every candidate, including ordinary and unscreened hypotheses: its agreement rates of 85%, 82% and 73% across strong, ordinary and unscreened strata expose its limited discrimination at the specified conditioning budget.

I add known-structure benchmarks against all-history Granger regression and PCMCI, a regime-switching comparison with Regime-PCMCI, and lattice comparisons with the official CaStLe implementation. Precision, recall, F1 and SHD are reported. The information supplied to each method is explicit, including true-state-mask comparators versus hidden-state learning. Component comparisons isolate own/full conditioning, OLS/HAC, diurnal adjustment and candidate support; they do not constitute a complete factorial ablation. Runtime and memory are now reported: the full three-stage analysis takes 118 seconds and peaks at 2.4 GB on the stated workstation. I position the contribution as a verification framework and retain failures under hidden confounding, rather than claiming a superior general identification algorithm.

Revised manuscript: Sections 2.2, 3.2, 3.5–3.7, 3.11, 4.2, 4.8 and 5.4; Tables 1, 2, 4 and 9; Figure 3; Supplementary Tables S1, S11 and S13; Supplementary Data S1.

## Response to Reviewer 3

### R3-1 · Contribution relative to existing methods

#### COMMENT

Regime-dependent causal discovery has already been studied by Regime-PCMCI, and recent work such as CaStLe addresses scalable space-time causal discovery through locality constraints in Earth-system data. The contribution here appears to lie mainly in combining physics-guided candidate restriction, manually defined atmospheric regimes, large-scale ERA5 analysis, and a layered validation chain. The authors should explain more precisely what new causal discovery capability is added beyond these prior approaches and include direct methodological comparisons.

#### RESPONSE

I have rewritten the contribution around a screen-confirm-replicate procedure: own-history screening, confirmation conditional on all observed regional histories, and frozen temporal evaluation with an unconfirmed-candidate control. I do not claim a new general causal-identification algorithm. The contribution is the explicit verification of which screened dependencies survive these checks, together with data-provenance and satellite-product tests.

I now compare directly with Regime-PCMCI on a two-state system and with the official CaStLe implementation on homogeneous and heterogeneous lattices. Regime-PCMCI achieves mean F1 0.826 versus 0.610 for pooled PCMCI and learns hidden states, a capability absent from the prescribed-regime analysis. CaStLe achieves F1 0.971-0.976 under homogeneity and 0.780-0.786 under heterogeneous structure; the latter does not establish that local estimators dominate its overall F1. Table 1 distinguishes their information sets and capabilities, and Table 9 reports recovery, runtime and memory comparisons where applicable. I retain each method's advantages and restrict the novelty claim to the combined confirmation, replication and negative-control workflow.

Revised manuscript: Sections 1, 2.2, 3.11, 4.8 and 5.3; Tables 1 and 9; Supplementary Table S13.

### R3-2 · Known-structure and full-candidate evaluation

#### COMMENT

The true directed graph of the real ERA5 data is unavailable. A method paper needs evaluation where the causal structure is known. PCMCI is applied only to 60 preselected top stable edges from 2019 to 2025, with 50 confirmed, so this is a selective confirmation rather than a full method comparison. Synthetic or semi-synthetic spatio-temporal benchmarks with known direction, lag, and graph structure should be added, together with precision, recall, F1, SHD, or related graph-recovery measures and comparisons with relevant causal discovery baselines.

#### RESPONSE

I have added six-variable benchmarks with known directed lagged graphs, 4,096 observations and 50 replicates per scenario, covering autocorrelation, observed and hidden common drivers, and regime switching. Precision, recall, F1, structural Hamming distance and false-discovery proportions are reported for own-history regression, all-history Granger regression and PCMCI. Under strong autocorrelation, own-history OLS has precision 0.086 and F1 0.158, whereas all-history regression and PCMCI have F1 about 0.97. With a hidden driver, all three approaches fail. Additional regime-switching and spatial-lattice benchmarks assess Regime-PCMCI and CaStLe.

I have also replaced the selected-edge PCMCI comparison with evaluation of all 2,574 ERA5 candidate-lag hypotheses. At its stated three-variable conditioning budget, PCMCI flags 85% of strong, 82% of ordinary and 73% of unscreened hypotheses, so agreement is not treated as ground-truth accuracy. The principal real-data evidence is now full-conditioning confirmation followed by retrospective held-out replication, with every unconfirmed hypothesis retained as a control. These simulation recovery scores and real-data replication rates answer different questions and are reported separately.

Revised manuscript: Sections 3.5-3.7, 3.11, 4.1-4.2 and 4.8; Tables 4 and 9; Figures 2-3; Supplementary Table S13.

### R3-3 · Dependence-aware uncertainty

#### COMMENT

The main estimator uses OLS nested F-tests with three target-history lags (as defined in Section 3.7, Eqs. (21)-(25)). However, FDR correction does not correct p-values that are themselves distorted by residual serial dependence. The block-bootstrap analysis in Section 3.10 uses only 20 replicates for stability selection. Residual autocorrelation should be diagnosed, and the authors should consider block-bootstrap, HAC, or other dependence-aware inference for the main edge tests.

#### RESPONSE

I agree that BH cannot repair distorted marginal p-values. I now use calendar-time Bartlett HAC uncertainty for the main coefficient tests, with bandwidth 64 six-hour steps and sensitivity values 32 and 128. Regime exclusions leave zero scores at their original times rather than compressing the calendar. The revised residual diagnostics show median screening autocorrelations of 0.263 at 24 hours and 0.215 at 72 hours; removing the diurnal cycle reduces, but does not eliminate, this dependence.

I also report 200 calendar-preserving bootstrap replicates at each of three block lengths for twelve fixed models. This replaces reliance on the earlier 20-replicate stability analysis, but does not establish selection-adjusted coverage. Known-null simulations provide the separate calibration assessment: in a heteroskedastic AR(4) setting, HAC empirical FDR is 0.270 at length 2,048 and 0.100 at 16,384. Consequently, I do not claim exact 5% graph-wide calibration. The revised interpretation combines explicitly stated nominal testing families with full conditioning, frozen prediction and held-out replication, while retaining the inference limitation.

Revised manuscript: Sections 3.5-3.7, 3.11, 4.1 and 5.4; Figure 2; Supplementary Table S8 and Supplementary Data S3.

### R3-4 · Outcome-dependent regime definitions

#### COMMENT

The regime construction can condition on the current outcome variable. High-humidity and high-cloud regimes are defined from contemporaneous humidity and cloud indices, and the regime mask is applied to the response time  $t$ . Thus, for relations such as  $H_{\{t-1\}} \rightarrow C_t$ , the high-cloud analysis conditions directly on a quantity derived from  $C_t$ . Matched sample sizes do not address this selection issue. Sensitivity analyses using lagged or otherwise externally defined regime indicators are needed.

#### RESPONSE

I now state explicitly that contemporaneous cloud and humidity masks can select on the response and that matched sample sizes do not remove this problem. I retain those graphs only as descriptive subsets. The sensitivity analysis uses discovery-fitted thresholds, antecedent indices at  $t-4$ , and an external Niño-3.4 state based on the last complete month before the earliest tested source.

I test source-by-state coefficient differences directly with state-specific intercepts and target-history effects and calendar-time HAC uncertainty. For cloud states, 173 concurrent and 124 antecedent interactions pass BH in discovery, compared with only 4 and 1 of 2,574 held-out tests. Humidity-state interactions retain limited support. External Niño-3.4 contrasts are small and control-sensitive: adding six antecedent atmospheric fields reverses 30 of 108 high-minus-middle and 24 of 108 low-minus-middle signs. I therefore withdraw the interpretation that cloud-state graph differences establish robust state-dependent coupling. Antecedent or external definitions reduce direct outcome selection, but they do not by themselves establish an exogenous regime or a complete confounding adjustment.

Revised manuscript: Sections 3.8, 4.4 and 5.1; Table 6; Figure 5; Supplementary Tables S6-S7.

### R3-5 · Directional moisture-transport diagnostics

#### COMMENT

Wind is reduced to scalar speed  $w$ , which contains no transport direction. However, cross-region  $W \rightarrow H$  edges are interpreted in terms of moisture transport. The authors should either incorporate vector wind information or narrow this interpretation.

#### RESPONSE

I have narrowed scalar wind-speed edges to conditional predictive relations and added directional diagnostics. Source wind and grid-cell moisture flux are projected onto the source-target great-circle bearing. These add held-out information beyond scalar speed in 331 and 320 of 396 models, respectively, but with small median gains. Conservative moisture-flux convergence and resolved flux summaries are also tested in two fixed Atlantic footprints; they improve humidity-change prediction by about 11-13%, while the cloud intervals include zero.

A separate pre-specified WHEC test isolates convergence of the wind-humidity anomaly product. Beyond the dense autoregression, local vector wind and linear convergence, it yields pooled midlatitude cloud-error reductions of 0.456% and 0.359% in the two held-out segments; remote-region and time-shifted controls fail the endpoint. This is limited predictive evidence for a resolved transport term. I do not interpret scalar-speed edges as directional fluxes, WHEC as a complete moisture budget, or the resulting dependencies as an identified transport-mediated causal chain. Vertical motion, subgrid transport and the differing vertical supports remain limitations.

Revised manuscript: Sections 3.2, 3.9-3.10, 4.5, 4.7 and 5.2-5.4; Figures 6 and 8; Table 8; Supplementary Tables S16-S17.

### R3-6 · Reverse direction and common drivers

#### COMMENT

The falsification results raise a substantive directionality issue. In Table 8, forward  $H \rightarrow C$  has a significant ratio of 0.740, whereas  $C \rightarrow H$  reaches 0.801, and future-to-past Humidity to Cloud Cover also remains strong. It indicates that the direction of the humidity-cloud relation is not cleanly identified under the current controls. The main interpretation should reflect this more directly, and additional conditioning on common atmospheric drivers would be useful.

#### RESPONSE

I agree that the original reverse and future-to-past results undermine direction claims from own-history screening. I retain the earlier falsification results as evidence of this limitation and do not interpret future-source prediction as backward causation. The principal confirmation model now conditions on all 264 regional histories, and symmetric candidates give both directions identical spatial opportunities. Additional sensitivity models include antecedent vertical pressure velocity at 500 and 700 hPa, 500-hPa geopotential, 700-hPa temperature, surface pressure and mean sea-level pressure.

With full conditioning, humidity-to-cloud has the larger absolute statistic in 55.2% of matched discovery comparisons, with modest asymmetry in both ERA5 held-out segments. Wind-to-humidity does not show consistent forward superiority; in the second segment the reverse is marginally stronger in the descriptive paired comparison. Replacing ERA5 cloud with CERES supports humidity-cloud dependence, but the

directional asymmetry is absent in the shorter 2023-2025 satellite segment. I therefore distinguish limited predictive asymmetry from causal orientation and do not claim that added controls eliminate the future-to-past diagnostic across the full graph. The six atmospheric fields are not a complete adjustment set, and the hidden-driver benchmarks show that full observed-history conditioning can still fail.

Revised manuscript: Sections 3.2, 3.6, 3.9, 4.3, 4.6 and 5.4; Tables 5 and 7; Figure 4; Supplementary Tables S7, S9 and S12-S13.

### R3-7 · Null calibration and predictive magnitude

#### COMMENT

The very high discovery rate needs a stricter null calibration. The main analysis reports 14274 significant tests out of 18018, or 0.792. More importantly, the distance-matched random graph still has a significant ratio of 0.743 in Table 6. This small separation suggests that spatial dependence accounts for much of the raw significance. Together with the very small median partial  $R^2$  values in Table 17, this study would benefit from an empirical-null comparison or an effect-size criterion in addition to  $q < 0.05$ .

#### RESPONSE

I have separated three questions: false discoveries under genuine no-effect simulations, specificity relative to spatial candidate references, and predictive effect size. Known-null experiments leave finite-sample calibration failures. For spatial specificity, 100 distance-stratified candidate sets give a mean crossing fraction of 0.813 versus 0.857 for the revised physical candidates; the 3,733 physical chains are not enriched relative to the null mean of 3,871. These full-record all-regime fractions use 2,574 hypotheses and should not be directly compared with the original 18,018-test fraction.

The main result now requires full-conditioning confirmation and held-out directional replication, with frozen predictive gains reported separately. Among replicated hypotheses, the median held-out partial  $R^2$  is 0.0029 and the median frozen relative MSE reduction is 0.27%; 416 of the 760 primary replicated hypotheses also pass the frozen-gain test. I retain these small magnitudes and the unconfirmed-candidate control rates rather than equating significance with physical importance. Positive predictive gain is an explicit additional diagnostic, not a post hoc universal physical-effect cutoff, and neither graph randomization nor replication proves nominal full-graph FDR.

Revised manuscript: Sections 3.5-3.7, 3.11, 4.1-4.3 and 5.4; Tables 4-5; Figures 2-4; Supplementary Table S14 and Supplementary Data S1.